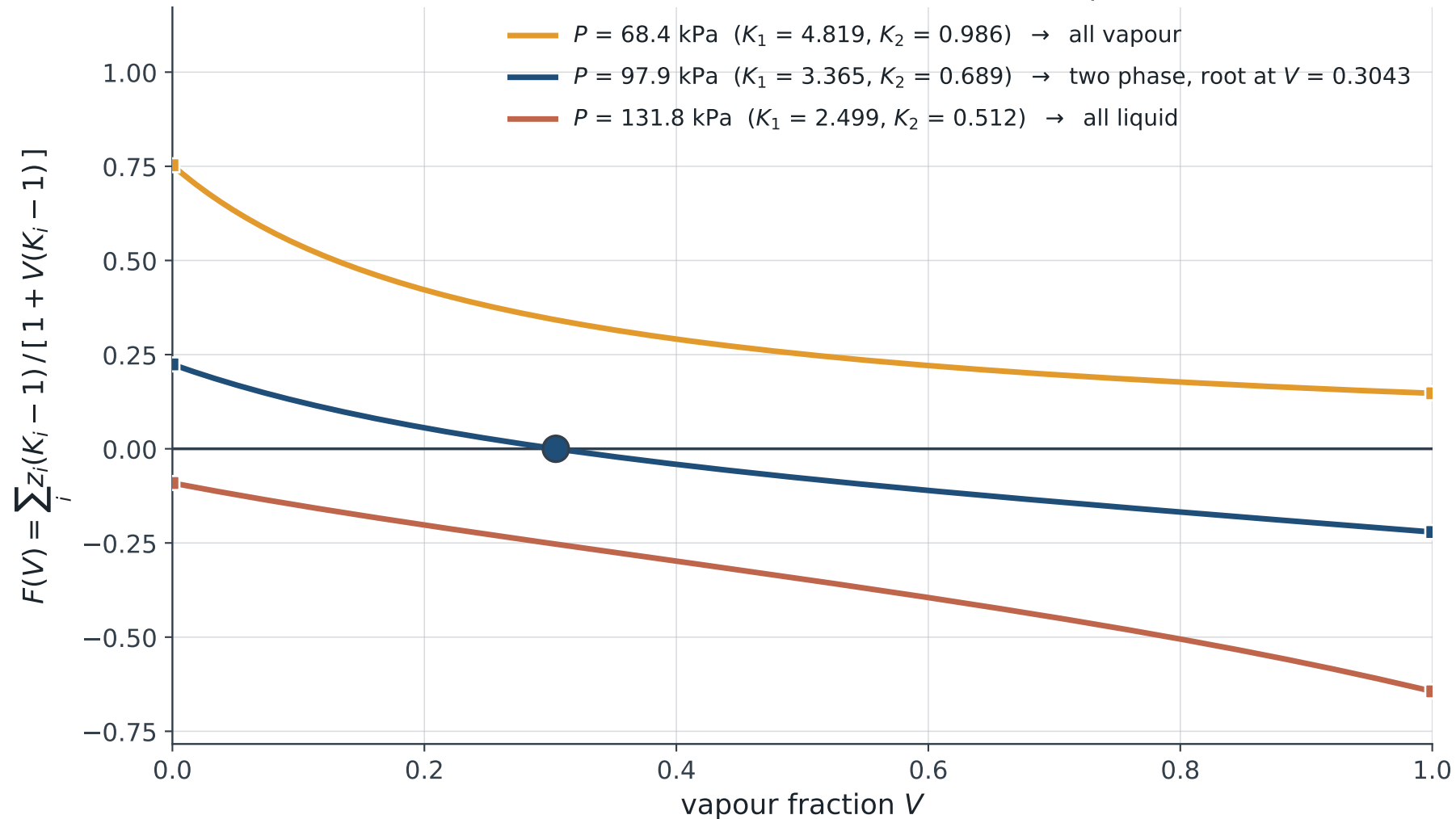


The Rachford-Rice function for one feed at three pressures



Ethanol (1) / water (2), Margules-2 with $A_{12} = 1.6, A_{21} = 0.9$ (illustrative, as in F4.2). Feed $z_1 = 0.2$ at $T = 360.00 \text{ K}$; bubble 119.83 kPa , dew 75.96 kPa .

Largest dF/dV over all three curves = -0.126 : F is strictly decreasing, so there is at most one root and the signs of $F(0)$ and $F(1)$ settle the phase state before any iteration begins.